

A Memoir on Cubic Surfaces.

By Professor CAYLEY.

[Continued from Vol. VI, pp. 259–366.]

Section I = 12. Article Nos. 38 to 46. (Continued.)

Double-sixers.

It has been mentioned that the number of double-sixers was = 36; these are as follows:

1, 2, 3, 4, 5, 6	Assumed primitive	
1', 2', 3', 4', 5', 6'		
1, 1', 23, 24, 25, 26	Like arrangements	15
2, 2', 13, 14, 15, 16		
1, 2, 3, 56, 46, 45	Like arrangements	20
23, 13, 12, 4, 5, 6		
	Total	36

where, if we take any column of two lines, we have the complete number 216 of pairs of non-intersecting lines (each line meets 10 lines, there are therefore $27 - 1 - 10 = 16$ lines which it does not meet; $27 \times 16 \div 2 = 216$).

Dr. Hart's Arrangement.

Dr. Hart arranges the 27 lines, cubically, thus:

A_1	B_1	C_1	a_1	b_1	c_1	α_1	β_1	γ_1
A_2	B_2	C_2	a_2	b_2	c_2	α_2	β_2	γ_2
A_3	B_3	C_3	a_3	b_3	c_3	α_3	β_3	γ_3

where letters of the same alphabet denote lines in the same plane, if only the letters are the same or the suffixes the same; thus A_1, A_2, A_3 lie in a plane $A_1A_2A_3$; A_1, B_1, C_1 lie in a plane $A_1B_1C_1$. Letters of different alphabets denote lines which meet according to the Table.

Equations of Planes.

Using the notation where $[12'] = w$, $[23'] = \theta$, etc., the equations of the planes are:

$$\begin{aligned}
W &= 0, & [12'] &= w \\
X &= 0, & [42'] &= y \\
Y &= 0, & [14'] &= z \\
Z &= 0, & [12'] &= w \\
\frac{X}{l} + \frac{Y}{m} + \frac{Z}{n} + W &= 0, & [41'] &= \xi \\
\frac{X}{l} + mY + \frac{Z}{n} + W &= 0, & [34'] &= \eta \\
\frac{X}{l} + \frac{Y}{m} + nZ + W &= 0, & [13.24.56] &= h \\
lX + mY + nZ + W &= 0, & [24'] &= \zeta \\
lX + \frac{Y}{m} + nZ + W &= 0, & [14.25.36] &= g \\
lX + mY + \frac{Z}{n} + W &= 0, & [43'] &= \xi \\
\frac{l(p-\alpha) + 2mn}{p+\alpha} X + \dots &= 0, & [52'] &= \psi \\
\frac{n(p-\alpha) + 2lm}{p+\alpha} Y + \dots &= 0, & [15'] &= z \\
\frac{m(p-\alpha) + 2nl}{p+\alpha} Z + \dots &= 0, & [12.36.45] &= \chi
\end{aligned}$$

Further Plane Equations.

Continuing the equations of the planes:

$$\begin{aligned}
-m \left\{ \frac{n(p-\alpha)}{\dots} \right\} X + mY + nZ + W &= 0, & [56'] &= l \\
lX - m \left\{ \frac{l(p-\alpha)}{\dots} \right\} Y + nZ + W &= 0, & [45'] &= m \\
lX + mY - n \left\{ \frac{m(p-\alpha)}{\dots} \right\} Z + W &= 0, & [64'] &= n \\
\frac{l(p-\alpha)}{\dots} X + mY + \frac{Z}{n} + W &= 0, & [15.26.34] &= l_1 \\
lX + \frac{2n}{\dots} Y + nZ + W &= 0, & [16.24.35] &= m_1 \\
lX + mY - \frac{m(p-\alpha)}{\dots} Z + W &= 0, & [14.25.36] &= n_1 \\
\frac{2n}{\dots} X + mY + \frac{Z}{n} + W &= 0, & [65'] &= l_1 \\
\frac{X}{l} + \frac{2n}{\dots} Y + nZ + W &= 0, & [46'] &= m_1 \\
\frac{X}{l} + mY - \frac{m(p-\alpha)}{\dots} Z + W &= 0, & [54'] &= n_1 \\
\frac{X}{l} + mY + \frac{Z}{n} + W &= 0, & [16.25.34] &= l_1 \\
\frac{2l}{\dots} X + mY + nZ + W &= 0, & [15.24.36] &= m_1 \\
lX - \frac{2l}{\dots} Y + nZ + W &= 0, & [14.26.35] &= n_1
\end{aligned}$$

Complex Plane Equations.

The remaining plane equations involve more complex coefficients:

$$-\frac{2l}{\dots}X + nY + mZ + \{mn(p - \alpha) - 2l(l^2 - m^2 - n^2)\}W = 0,$$

$$[36' = p_1]$$

$$mX - \frac{2m}{\dots}Y + lZ - \frac{l}{\dots}\{m(1 - n^2)(p - \alpha) - 2nl\}W = 0,$$

$$[25' = q_1]$$

$$nX + lY - \frac{2n}{\dots}Z - \frac{l}{\dots}\{n(1 - l^2 - m^2)(p - \alpha) - 2lm\}W = 0,$$

$$[13.26.45 = r_1]$$

Coordinates of the 27 Lines.

The coordinates of the 27 lines are then found to be as follows:

(a) (b) (c)

$$\begin{array}{lllll} 1 : & 1, & 1, & 1, & 1 - k \left\{ \frac{l}{m} \right\} \left\{ \frac{m}{n} \right\} \dots \\ 2 : & 1, & -k \left\{ \frac{l}{m} \right\} \left\{ \frac{m}{n} \right\}, & 1, & 1 \dots \\ 3 : & -k(l-1)(m-1), & -k(l-1)(m-1), & 1, & \dots \end{array}$$

(f) (g) (h)

$$\begin{array}{lllll} (12 = a_1) : & -n, & m, & 1, & -l \dots \\ (1' = b_2) : & 1, & -l, & n, & m \dots \\ (23 = b_1) : & -m, & l, & 1, & n \dots \\ (3' = c_1) : & 1, & 1, & -l, & m \dots \end{array}$$

The remaining line coordinates follow similar patterns with appropriate permutations of the parameters l, m, n and their combinations.

Equation of the Cubic Surface.

We have $X = 0, Y = 0, Z = 0, W = 0$ for the equations of the planes $(12.34.56 = x), (42' = y), (14' = z), (12' = w)$; and representing by $\xi = lX + \frac{Y}{m} + \frac{Z}{n} + W = 0$ the equation of any other plane

($41' = \xi$), the equation of the cubic surface may be presented in the several forms:

$$\begin{aligned}
&= Fgg_1 + k\eta\zeta ZX, \\
&= Fhh_1 + k\zeta\theta XY, \\
&= W\eta\eta_1 + k\theta\iota YZ, \\
&= W\zeta\zeta_1 + k\iota\kappa ZX, \\
&= W\theta\theta_1 + k\kappa\lambda XY, \\
&= W\iota\iota_1 + k\lambda\mu YZ, \\
&= W\kappa\kappa_1 + k\mu\nu ZX, \\
&= W\lambda\lambda_1 + k\nu\xi XY, \\
&= W\mu\mu_1 + k\xi\eta YZ, \\
&= W\nu\nu_1 + k\xi\eta\zeta, \\
&= W\xi\xi_1 + k\eta\zeta\iota, \\
&= W\pi\pi_1 + k\xi\eta z, \\
&= W\rho\rho_1 + k\zeta\theta z, \\
&= W\sigma\sigma_1 + k\xi\iota x.
\end{aligned}$$

Section II = $12-C_5$. Article Nos. 47 to 59.

The Reciprocal Surface.

It may be remarked that the system of lines and planes is at once deduced from that belonging to I = 12, by supposing that in the double-sixer the lines 1, 2, 3, 4, 5, 6 become the rays of a cubic node C_5 . The reciprocal of a double-sixer is a double-sixer. Hence the 27 lines of the reciprocal surface may be (and that in 36 different ways) represented by:

$$\begin{array}{cccccc}
1, & 2, & 3, & 4, & 5, & 6 \\
1', & 2', & 3', & 4', & 5', & 6' \\
12, & 13, & \dots & 56,
\end{array}$$

where 12 is now the line joining the points $12'$ and $1'2$; and so for the other lines. The lines 12, 34, 56 meet in a point 12.34.56; the 30 points $12', 1'2, \dots, 56', 5'6$, and the fifteen points 12.34.56 make up the 45 points t' .

The above equation, $S^3 - T^2 = 0$, shows that the cuspidal curve is a complete intersection 6×4 ; $c' = 24$.

Equation $W(a, b, c, f, g, h \setminus X, Y, Z)^2 + 2kXYZ = 0$.

The system of lines and planes is at once deduced from that belonging to II = $12 - C_5$, by supposing the tangent cone to reduce itself to the pair of biplanes.

The Diagram.

Lines.	
12	
13	
14	
15	
16	
23	Biradial planes, through each pair of rays. $15 \times 2 = 30$
24	
25	
26	
34	
35	
36	
45	
46	
56	
12.34.56	
12.35.46	
12.36.45	Planes each containing three non-intersecting lines. $15 \times 1 = 15$
13.24.56	
13.25.46	
13.26.45	
14.23.56	
14.25.36	
14.26.35	
15.23.46	
15.24.36	
15.26.34	
16.23.45	
16.24.35	
16.25.34	
30	45

Equations of Planes.

Putting the equation of the surface in the form:

$$W \{1, 1, 1, 1 \setminus X, Y, Z\}^2 + \frac{2}{P} XYZ = 0,$$

where for shortness:

$$\lambda = mn - l, \quad \mu = nl - m, \quad \nu = lm - n, \quad \delta = lmn - 1, \quad \rho = lmn,$$

then taking $X = 0$ as the equation of the plane [12], $Y = 0$ as that of the plane [34], $Z = 0$ as that of the plane [56], the equations of the 30 distinct planes are found to be:

$$\begin{aligned}
X &= 0, & [12] \\
Y &= 0, & [34] \\
Z &= 0, & [56] \\
mX + lY + Z &= 0, & [23] \\
m^{-1}X + Y + Z &= 0, & [24] \\
X + nY + mZ &= 0, & [45] \\
mX + l^{-1}Y + Z &= 0, & [13] \\
m^{-1}X + l^{-1}Y + Z &= 0, & [14] \\
X + n^{-1}Y + mZ &= 0, & [46]
\end{aligned}$$

Coordinates of the 21 Distinct Lines.

And the coordinates of the 21 distinct lines are:

(a) (b) (c) (f) (g) (h) whence equations may be taken to be:

1 :	$l,$	$-1,$	$1,$	(1)	$X = 0, \quad Y + lZ = 0$
2 :	$1,$	$-1,$	$1,$	(2)	$Y = 0, \quad Z + mX = 0$
3 :	$n,$	$-1,$	$1,$	(3)	$Z = 0, \quad X + nY = 0$
4 :	$1,$	$-1,$	$1,$	(4)	$X = 0, \quad Y + nZ = 0$
5 :	$1,$	$-1,$	$1,$	(5)	$Y = 0, \quad Z + lX = 0$
6 :	$m,$	$n,$	$1,$	(6)	$Z = 0, \quad X + m^{-1}Y = 0$

And for the remaining lines:

(45) :	$\frac{1}{n},$	$\frac{1}{m},$	$1,$	$\frac{\alpha}{\gamma},$	$\frac{1}{\gamma},$	$\frac{1}{\beta}$
(16) :	$\frac{1}{n},$	$\frac{1}{l},$	$1,$	$\frac{\gamma}{\beta},$	$\frac{1}{\beta},$	$\frac{1}{\alpha}$
(23) :	$\frac{1}{m},$	$l,$	$1,$	$\frac{\beta}{\alpha},$	$\frac{1}{\alpha},$	$\frac{1}{\gamma}$
(46) :	$n,$	$m,$	$1,$	$\frac{1}{\gamma},$	$\frac{\alpha}{\gamma},$	$\frac{1}{\beta}$
(26) :	$l,$	$m,$	$1,$	$\frac{1}{\beta},$	$\frac{\gamma}{\beta},$	$\frac{1}{\alpha}$

The Hessian Surface.

Resuming the equation $W(a, b, c, f, g, h \setminus X, Y, Z)^2 + 2kXYZ = 0$, the equation of the Hessian surface is found to be:

$$\begin{aligned}
& kW^2(a, b, c, f, g, h \setminus X, Y, Z)^2 \\
& + 2kW\{(a, b, c, f, g, h \setminus X, Y, Z)^2(FX + GY + HZ) - k^2XYZ\} \\
& - k\{a^2X^4 + b^2Y^4 + c^2Z^4 - 2bcY^2Z^2 - 2caZ^2X^2 - 2abX^2Y^2 \\
& - 4XYZ[(af + gh)X + (bg + hf)Y + (ch + fg)Z]\} = 0,
\end{aligned}$$

where

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

and

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh.$$

Invariants and Reciprocal Surface.

Write as before (A, B, C, F, G, H) for the inverse coefficients ($A = bc - f^2$, &c.), and $K = abc - af^2 - bg^2 - ch^2 + 2fgh$; and moreover:

$$\begin{aligned}
\Phi &= (A, B, C, F, G, H \setminus x, y, z)^2, \\
P &= Ax + Hy + Gz, \\
Q &= Hx + By + Fz, \\
R &= Gx + Fy + Cz, \\
t &= fx + gy + hz, \\
U &= afyz + bgzx + chxy, \\
V &= 2Kxyz - aPyz - bQzx - cRxy \\
&= -aHy^2z - bFz^2x - cGx^2y - aGyz^2 - bEzx^2 - cFxy^2 \\
&\quad + (-abc - ap^2 - bg^2 - ch^2 + 4fgh)xyz, \\
W &= (A, B, C, F, G, H \setminus ayz, bzx, cxy)^2, \\
L &= kW - 2ktw - \Phi, \\
M &= kwU + V, \\
N &= 2kabcxyzw + W.
\end{aligned}$$

Then the invariants of the ternary cubic are:

$$\begin{aligned}
S &= L^2 - 12kwM, \\
T &= L^3 - 18kwLM - 54k^2w^2N,
\end{aligned}$$

and the required equation of the reciprocal surface is:

$$\frac{1}{k^2w^2} [(L^3 - 18kwLM - 54k^2w^2N)^2 - (L^2 - 12kwM)^3] = 0,$$

viz. this is:

$$0 = DN = (kW - 2ktw - \Phi)^2(2kabcxyzw + W) + L^2M^2 + \dots$$

Section III = 12- B_3 . Article Nos. 60 to 72.

Equation $2W(X + Y + Z)(lX + mY + nZ) + 2kXYZ = 0$.

The system of lines and planes is at once deduced from that belonging to $\text{II} = 12 - C_5$, by supposing the tangent cone to reduce itself to the pair of biplanes: three of the planes (α) of $\text{II} = 12 - C_5$ thus coming to coincide with the one biplane, and three of them with the other biplane.

The Diagram.**Lines.**

$$\text{III} = 12 - B_3$$

1

2

1'

2'

12 Biplanes containing rays 1, 2 and 1', 2' respectively. $2 \times 12 = 24$

11'

22' Plane touching along edge and containing the transversal. $1 \times 8 = 8$

12'

21' Biradial planes each containing a ray of the one and a ray of the other biplane. $4 \times 4 = 16$

11'.23'

12'.31' Planes each through the transversal. $2 \times 1 = 2$

45

46

56

3

4

5

6

3'

4'

5'

6'

Equations of Planes.

Taking $X + Y + Z = 0$ for the biplane that contains the rays 1, 2, 3, and $lX + mY + nZ = 0$ for that which contains the rays 4, 5, 6, we may take $X = 0$, $Y = 0$, $Z = 0$ for the equations of the planes [14], [25], [36] respectively; and then writing for shortness:

$$m - n = \lambda, \quad n - l = \mu, \quad l - m = \nu,$$

and assuming, as we may do, $k = \sqrt{\lambda\mu\nu}$, so that the equation of the surface is:

$$W(X + Y + Z)(lX + mY + nZ) + (m - n)(n - l)(l - m)XYZ = 0,$$

the equations of the 17 distinct planes are:

$$X = 0, \quad [14]$$

$$Y = 0, \quad [25]$$

$$Z = 0, \quad [36]$$

$$X + Y + Z = 0, \quad [123]$$

$$lX + mY + nZ = 0, \quad [456]$$

Coordinates of the 15 Distinct Lines.

And the coordinates of the fifteen distinct lines are:

(a) (b) (c) (f) (g) (h) whence equations may be written:

(1) :	- 1,	1,	1,	(1) $X = 0,$	$Y + Z = 0$
(2) :	1,	- 1,	1,	(2) $Y = 0,$	$Z + X = 0$
(3) :	1,	1,	- 1,	(3) $Z = 0,$	$X + Y = 0$
(1') :	- n ,	m ,	1,	(4) $X = 0,$	$mY + nZ = 0$
(2') :	n ,	- l ,	1,	(5) $Y = 0,$	$nZ + lX = 0$
(3') :	l ,	m ,	- 1,	(6) $Z = 0,$	$lX + mY = 0$
(14) :	1,	- 1,	1,	(14) $X = 0,$	$W = 0$
(25) :	1,	1,	- 1,	(25) $Y = 0,$	$W = 0$
(36) :	- 1,	1,	1,	(36) $Z = 0,$	$W = 0$

And the remaining lines:

(15) :	l ,	n ,	n ,	$n^2\nu - n\lambda\mu$
(16) :	l ,	m ,	m ,	$m^2\mu - lm\nu$
(26) :	l ,	m ,	l ,	$l^2\lambda - lm\mu$
(24) :	n ,	m ,	n ,	$n^2\nu - mn\lambda$
(34) :	m ,	m ,	n ,	$m^2\mu - mn\nu$
(35) :	l ,	l ,	n ,	$l^2\lambda - ln\nu$

The Rays and Facultative Lines.

The rays are not, the mere lines are, facultative; hence $b' = p' = 6$; $t' = 6$.

The Hessian Surface.

The equation of the Hessian surface is:

$$\begin{aligned}
 & -W(X + Y + Z)(lX + mY + nZ)(\nu X + \lambda Y + \mu Z) \\
 & -k(l^2X^2 + m^2Y^2 + n^2Z^2 - 2mnYZ - 2nlZX - 2lmXY) = 0.
 \end{aligned}$$

Section IV = 12- B_4 . Article Nos. 73 to 84.

Equation $ZW(e, X - Y)(e_2X - Y)(e_3X - Y)(e_4X - Y) + 2kXYZ = 0$.

Writing $Z(a, b, c, d \setminus X, Y)^2 \cdot d_1X - Y \cdot d_2X - Y \cdot d_3X - Y \cdot d_4X - Y$, the 20 planes are:

$$\begin{aligned}
 Z &= 0, & [0] \\
 X - f_1Y &= 0, & [11'] \\
 X - f_2Y &= 0, & [22'] \\
 X - f_3Y &= 0, & [33'] \\
 X - f_4Y &= 0, & [44'] \\
 Z - (f_1 + f_2)Y - f_1f_2Z &= 0, & [12] \\
 Z - (f_1 + f_3)Y - f_1f_3Z &= 0, & [13] \\
 Z - (f_1 + f_4)Y - f_1f_4Z &= 0, & [14] \\
 Z - (f_2 + f_3)Y - f_2f_3Z &= 0, & [23] \\
 Z - (f_2 + f_4)Y - f_2f_4Z &= 0, & [24] \\
 Z - (f_3 + f_4)Y - f_3f_4Z &= 0, & [34] \\
 Z - (f_1 + f_2)Y - f_1f_2Y &= 0, & [1'2'] \\
 Z - (f_1 + f_3)Y - f_1f_3Y &= 0, & [1'3'] \\
 Z - (f_1 + f_4)Y - f_1f_4Y &= 0, & [1'4'] \\
 Z - (f_2 + f_3)Y - f_2f_3Y &= 0, & [2'3'] \\
 Z - (f_2 + f_4)Y - f_2f_4Y &= 0, & [2'4'] \\
 Z - (f_3 + f_4)Y - f_3f_4Y &= 0, & [3'4']
 \end{aligned}$$

and the planes through the nodes:

$$\begin{aligned}
 (f_1 + f_2 + f_3 + f_4)Z + \cdots &= 0, & [12.34] \\
 (f_1 + f_3 + f_2 + f_4)Z + \cdots &= 0, & [13.24] \\
 (f_1 + f_4 + f_2 + f_3)Z + \cdots &= 0, & [14.23]
 \end{aligned}$$

The 16 Lines.

And the 16 lines are:

(a) (b) (c) (f) (g) (h) whence equations may be written:

$$\begin{aligned}
 (0) : & \quad 1, & 0, & 0, & x = 0, \quad Y = 0 \\
 (5) : & \quad 0, & -\gamma, & d, & X = 0, \quad dY + \gamma Z + dW = 0 \\
 (1) : & \quad 1, & e_1, & 0, & e_1X - Y = 0, \quad Z = 0
 \end{aligned}$$

and so for the other lines with appropriate substitutions of e_2, e_3, e_4 .

Section V = 12- B_3 . Article Nos. 85 to 94.

Equation $WXZ + (X + Z)(Y^2 - aX^2 - bZ^2) = 0$.

The diagram of the lines and planes is:

Lines.

$$V = 12 - B_3$$

1

2

1'

2' Biplanes containing rays 1, 2 and 1', 2' respectively. $2 \times 12 = 24$

12

11' Plane touching along edge and containing the transversal. $1 \times 8 = 8$

22'

12' Biradial planes each containing a ray of the one and a ray of the other biplane. $4 \times 4 = 16$

21'

11'.23' Planes each through the transversal. $2 \times 1 = 2$

12'.31'

3

4

5

6

3'

4'

5'

6'

The Planes.

The planes are:

$$Z = 0, \quad [12]$$

$$X = 0, \quad [1'2']$$

$$Y + \sqrt{a}X + \sqrt{b}Z = 0, \quad [11']$$

$$Y - \sqrt{a}X - \sqrt{b}Z = 0, \quad [22']$$

$$Y + \sqrt{a}X - \sqrt{b}Z = 0, \quad [12']$$

$$Y - \sqrt{a}X + \sqrt{b}Z = 0, \quad [21']$$

with other planes determined by the appropriate combinations.

Section VI = 12- B_4 . Article Nos. 95 to 107.**Equation** $Z(X - Y)(\theta_1 X - Y)(\theta_2 X - Y)(\theta_3 X - Y)(\theta_4 X - Y) = 0$.Writing $(a, b, c, d \setminus Z, Y)^2 \cdot d_1 X - Y \cdot e_1 X - Y \cdot e_2 X - Y \cdot e_3 X - Y \cdot e_4 X - Y$, the planes are:

$$\begin{aligned}
X &= 0, & [0] \\
Z &= 0, & [00] \\
\theta_1 X - Y &= 0, & [22'] \\
\theta_2 X - Y &= 0, & [33'] \\
\theta_3 X - Y &= 0, & [44'] \\
\theta_4 X - Y &= 0, & [55'] \\
d(e_1 X - Y) - Z &= 0, & [12] \\
d(e_2 X - Y) - Z &= 0, & [13] \\
d(e_3 X - Y) - Z &= 0, & [14] \\
X\theta_1 - Y(\theta_2 + \theta_3) - W &= 0, & [2'3'] \\
X\theta_2 - Y(\theta_1 + \theta_4) - W &= 0, & [2'4'] \\
X\theta_3 - Y(\theta_1 + \theta_4) - W &= 0, & [3'4']
\end{aligned}$$

The Lines.

And the lines are:

(a) (b) (c) (f) (g) (h) equations may be written:

(0) :	1,	0,	0,	$X = 0, \quad Y = 0$
(1) :	$-1,$	$d,$	$1,$	$X = 0, \quad dY + Z = 0$
(2) :	$1,$	$e_1,$	$1,$	$e_1 X - Y = 0, \quad Z = 0$
(3) :	$1,$	$e_2,$	$1,$	$e_2 X - Y = 0, \quad Z = 0$
(4) :	$1,$	$e_3,$	$1,$	$e_3 X - Y = 0, \quad Z = 0$

and for the remaining lines, the coordinate expressions are more conveniently written in the symmetrical form.

The mere lines are facultative; hence $b' = p' = t' = \dots$ **Section VII = 12- $2C_2$. Article Nos. 108 to 116.****Equation** $Y^2 + Y(X + Z + W) + 4\phi XZW = 0$.Each of the lines $(X - Y = 0, Y + Z = 0)$ and $(X + W = 0, Y - Z = 0)$ is a double tangent of the spinode octic; in fact for the first of these lines we have:

$$16\theta^4 + 8\theta^2 + 16\phi\theta^2 + 5 = 0, \quad 16\theta^4 + 16\theta^2 + 4\phi = 0,$$

that is:

$$(2\theta^2 + 1)(8\theta^2 - 4\theta + 5) = 0, \quad 4\theta(2\theta^2 + 1) = 0,$$

so that the line touches at the two points given by $2\theta^2 + 1 = 0$; and similarly the other line touches at the two points given by $2\theta^2 - 1 = 0$.The edge $X = 0, Z = 0$ has apparently a higher contact with the spinode octic, viz. the equations $X = 0, Z = 0$ are satisfied by $\theta = \infty$ twice, $\theta = 0$ five times; but it must be reckoned only as a double tangent. Hence $\beta' = 2 \cdot 2 + 2 = 6$.

Reciprocal Surface.

The equation is obtained by equating to zero the discriminant of the binary quartic:

$$Z^2(yZ - wX)^2 + 4wZ^2(wZ^2 + zZX + xX^2),$$

viz. calling $t = \dots$

Section VIII = $12-3C_2$. Article Nos. 117 to 125.

Equation $Y^2 + Y(X + Z + W) + 4\phi XZW = 0$.

The diagram of the lines and planes is:

Lines.

VIII = $12 - 3C_2$

1

7 Planes each touching along an axis and containing the corresponding transversal. $3 \times 2 = 6$

8

9

12

34 Biradial planes each containing two rays of the same node. $3 \times 2 = 6$

56

13

24 Planes each containing an axis, and two rays through the terminal nodes respectively. $6 \times 4 = 24$

16

26

46

35

789 Plane through the three axes. $1 \times 8 = 8$

789 Plane through the three transversals. $1 \times 1 = 1$

14

45

[End of Memoir on Cubic Surfaces, pp. 401-450 of Vol. VI.]